Airbnb Price Prediction Report - Edinburgh

Generated on 2026-08-25 by the automated pipeline (`make all`).

Executive summary

The pipeline trained and compared Random Forest, XGBoost and TabPFN on Airbnb listing data for **Edinburgh** (3447 listings). The **xgboost** model was selected for final evaluation on a held-out test set and achieved: **- RMSE: $162 - MAE: $68 - R²: 0.64**

Data

- Source: InsideAirbnb (latest available snapshot). See: https://insideairbnb.com/get-the-data/
- City: Edinburgh · 3447 listings after cleaning

EDA

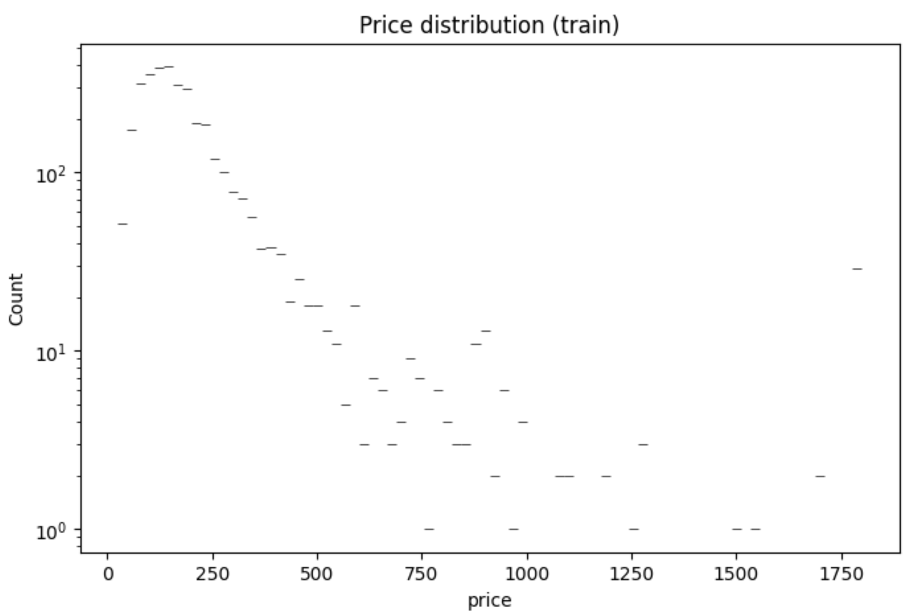
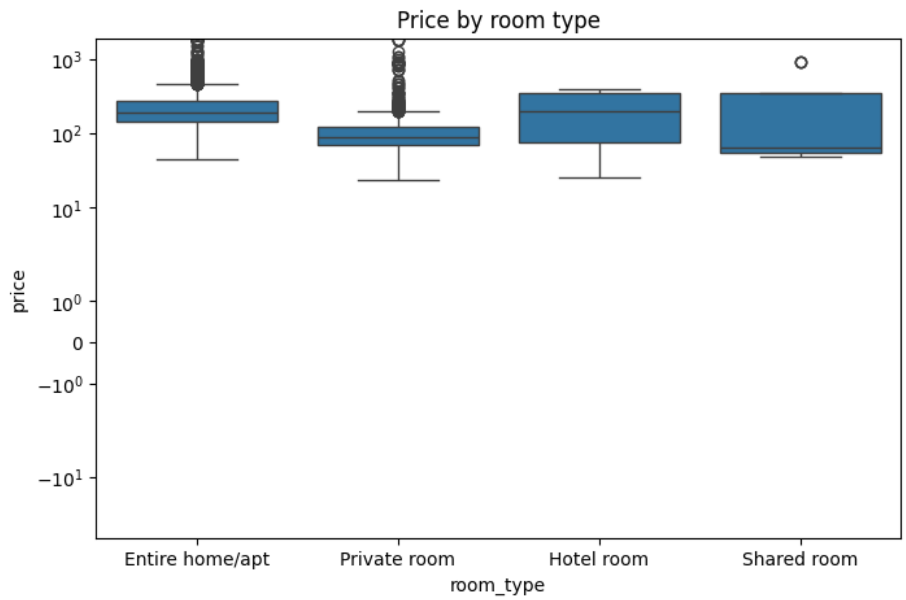
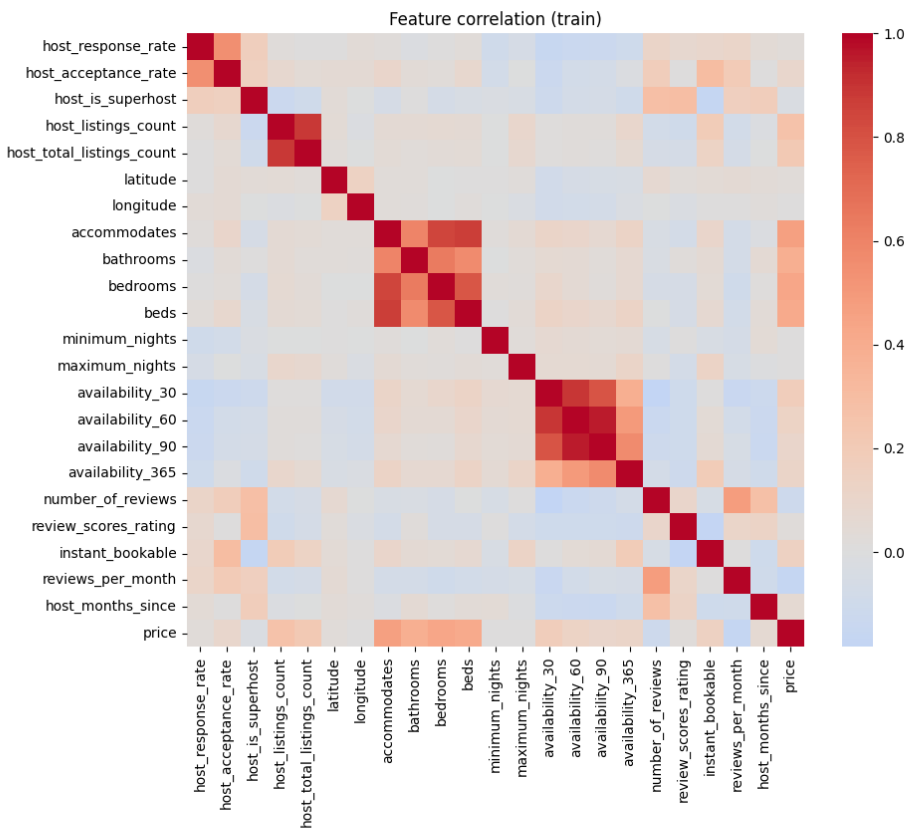


Figure 1: price_dist

- Median price: **$160** (mean $215)


Price by room type
price
room_type
Entire home/apt
Private room
Hotel room
Shared room


Feature correlation (train)
host_response_rate
host_acceptance_rate
host_is_superhost
host_listings_count
host_total_listings_count
latitude
longitude
accommodates
bathrooms
bedrooms
beds
minimum_nights
maximum_nights
availability_30
availability_60
availability_90
availability_365
number_of_reviews
review_scores_rating
instant_bookable
reviews_per_month
host_months_since
price

Models & Results

Model	Validation RMSE
Random Forest	$137
XGBoost	$123
TabPFN (capped sample, not directly comparable)	$122

Holdout results (best model): see `artifacts/edinburgh/holdout_results.csv`.

Top features

Feature	Importance
property_type_Private room in townhouse	0.083
host_neighbourhood_New Town	0.078
host_neighbourhood_Cannonmills	0.071
bedrooms	0.063
host_neighbourhood_Fasach	0.057
host_neighbourhood_Laulerie	0.042
accommodates	0.037
availability_90	0.034
bathrooms	0.034
host_listings_count	0.028

Recommendations

- The best-performing model was **xgboost** (holdout RMSE $162, MAE $68, R^2 0.64).
- On validation, **XGBoost** beat Random Forest by 10% on RMSE ($123 vs $137). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **property_type_Private room in townhouse, host_neighbourhood_New Town, host_neighbourhood_Cannonmills** — i.e. property_type_Private room in townhouse (the property type (apartment, house, etc.) (specifically Private room in townhouse)); host_neighbourhood_New Town (the host's neighbourhood (specifically New Town)); host_neighbourhood_Cannonmills (the host's neighbourhood (specifically Cannonmills)). Hosts can use these levers when setting nightly rates.
- Accuracy caveat: the holdout RMSE of $162 is **101% of the median listing price ($160)**, so predictions for typical listings are expected to be off by roughly ±$162.

- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Automated pipeline: fetch → preprocess → EDA → train → evaluate → report. Code in `src/`, config in `configs/base.yaml`.